

2026 Eclipse Shadow Band Citizen Science Event

Citizen Science Goals

- Collect additional scientific data to refine tools developed for shadow band analysis.
- Explore how altitude and proximity to the eclipse centerline affect eclipse shadow band observations.
- Have fun! Eclipse shadow bands are a rare and fascinating phenomenon to experience.

Safety Reminder

- Never look directly at the Sun during an eclipse without proper eye protection.

What Are Eclipse Shadow Bands?

- Eclipse shadow bands are a phenomenon that occurs immediately before and after totality during a solar eclipse.
- They appear as subtle, fast-moving light and dark bands caused by atmospheric turbulence affecting incoming sunlight.
- Although visible to the naked eye, shadow bands are difficult to record because of their low contrast.
- Tools were developed for extracting eclipse shadow band features from recorded video data - see '[Image Processing Framework for Eclipse Shadow Band Analysis](#)' for additional details

What You Need

- 1 tape measure or other means to measure distance
- 1 smartphone with video recording capability, GPS, and compass applications.
 - **Warning: Be sure your phone is fully charged! This experiment will use up its battery. Have a plan in case your phone runs out of battery.**
- 1 phone stand or tripod
- 1 white cloth sheet (white bed sheet or equivalent)
- 1 roll of packing tape (clear Scotch Heavy Duty shipping tape or equivalent)
- 5 small to medium cardboard boxes (16"x20"x15' or equivalent)



Step 1: Select Your Observation Location

- Choose a location within the path of totality that has a clear view of the northwest horizon.
- For this study, preferred locations will explore the effect of altitude and center line proximity on band effects
 - **Experiment Location 1: Along Center Line**
 - A location that is exactly along Eclipse center line
 - Examples: La Pola, Spain or Salas, Spain or equivalent
 - **Experiment Location 2: Edge of Center Line**
 - A location that is far as possible from Eclipse center line but still within totality
 - Example: Bilbao, Spain or Zamora, Spain or equivalent
 - **Experiment Location 3: High Altitude**
 - A location that is somewhere high in altitude, like on top of a mountain
 - Example: Pico San Millán, Spain or Moncayo, Spain or equivalent

Step 2: Arrive at Observation Location

- **Warning: Be sure your phone is fully charged! This experiment will use up its battery. Have a plan in case your phone runs out of battery.**
- Arrive at location a few hours before totality to give yourself plenty of time to setup and scout out a good location
- Find a location that has a clear view of the sun during the eclipse which will be near horizon
- The sun will be close to sunset during totality, so make sure the view to the Northwest is unobstructed
- A good location might be on top of a hill away from trees or buildings

Step 3: Assemble Board at Observation Location

- Attach four cardboard boxes together with overlap and tape to form a large board.



- Stretch white sheet over assembled board and secure with tape on the back.



- Assemble fifth box and place weight inside, like a bag. Prop board against fifth box to be vertical.



- Secure the overall structure firmly so it does not move in the wind.
- Maintain straight edges and right angles around the sheet or board to improve post-processing accuracy.
- Orient the surface roughly toward the Sun in the northwest horizon.
- Place camera stand in front of board to the side so that the shadow does not fall on board



Step 4: Record Setup Information

- Record width and height of the board
- Record your GPS latitude & longitude
- Record angle of the board's front surface



- Record compass orientation of the board's front surface. It should be in the Northwest direction.



Step 5: Setup Camera

- **Warning: Be sure your phone is fully charged! This experiment will use up its battery. Have a plan in case your phone runs out of battery.**
- Place the phone on a stable stand facing the board but slightly off to the side to avoid any shadows from the camera stand appearing on the board.
- Position the phone as close to the board as possible but far enough so that all four corners of the board are recorded.
- The post process image processing techniques can account for the camera being slightly off centered both laterally and vertically. It is more important to have a stable camera and board that will not move due to wind or other disturbances.
- Ensure the camera remains completely stationary throughout the recording.
- Use highest resolution available (HD)
- Use 60 frames per second (fps) if available. 30 fps is okay, but 60 is preferred.
- Turn off auto-focus
- Do not change camera settings once recording has begun.
- Do not move camera once recording has begun.

Step 6: Record Control Samples

- Record a 1-minute video sample prior to C1 (first contact). The exact time of C1 depends on your location. This should be about 90-minutes prior to totality. Note time of recording.
- Record a 1-minute sample after C1 (first contact) but before C2 (start of totality). This should be about 30 minutes prior to totality. Note time of recording.
- Use these control sample recordings to test your camera setup. If you mess up recording a control sample, start over.

Step 7: Record Shadow Bands

- Begin continuous recording 10 minutes before totality (C2).
- Continue recording through totality and until at least 10 minutes after totality ends (C3).
- Do not touch the camera or the sheet during recording.
- Keep all equipment stable throughout the observation period.
- Shadow bands typically appear a few minutes before and after totality.
- Watch for faint, shimmering light and dark bands.
- They may resemble ripples, waves, or heat-shimmer patterns.
- The bands can move rapidly across the surface.

Step 8: Share Your Data

- Share your video recordings with jconti@alum.mit.edu
- You can optionally analyze the data yourself using the available GitHub project.